

# Alexis DHERMY

## Applicant for M.S. Program in Artificial Intelligence – South Korea (Spring 2027)

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## EDUCATION

- Sep 2021 – **Master of Science in Engineering (French *Diplôme d'Ingénieur*)** | EPF Engineering School, Cachan, France  
Aug 2026 *Major: Intelligent Systems and Digital Technologies – AI & Cloud Computing Track*
- GPA: 15.87/20 | Rank: Top 3% of cohort – 16/276 (Y4), 6/206 (Y3), 4/178 (Y2), 4/217 (Y1)
  - 2-year preparatory cycle (Mathematics & Physics), 1-year general engineering, 2-year specialization (CS & AI)
  - Thesis: LLM-Based RAG Architecture for Enterprise Systems (*Viridien*) – Advisor: Prof. Zehira Haddad
  - Relevant Coursework: Generative AI, Deep Learning, Machine Learning, Probability & Statistics, Linear Algebra
- Sep 2024 – **Exchange Semester in Artificial Intelligence** | Sungkyunkwan University (SKKU), Seoul, South Korea  
Dec 2024 *College of Computing and Informatics – School of Convergence*
- GPA: 4.10/4.5
  - Relevant Coursework: Reinforcement Learning, Natural Language Processing, Large Language Models, AI Search Algorithms, Data Science

## HONORS & AWARDS

- **National Engineering Entrance Merit Distinction** (*Concours Avenir – Grand Classé*): Selected in the top tier of 10,000+ candidates in one of France's most competitive post-high school engineering entrance exams; earning full exam exemption.

## RESEARCH INTERESTS

- LLMs, Reasoning Models, AI Agents, Efficient Inference
- Reinforcement Learning, Embodied AI, Robotics, World Models

## ACADEMIC PROJECTS

- Nov 2024 – **Music Recommendation Using NLP** ([Code & Report](#)) | Sungkyunkwan University (SKKU)  
Dec 2024
- Developed an end-to-end NLP pipeline (NLTK, SpaCy, BERT) to perform tokenization, POS tagging, lemmatization, and sentiment analysis on top Spotify tracks across France, Spain, and Korea.
  - Built a cross-cultural recommendation engine that maps semantic and emotional features to recommend French/Spanish tracks based on a target Korean song.
- Nov 2024 – **Teaching a Robot to Walk with Reinforcement Learning** ([Presentation](#)) | Sungkyunkwan University (SKKU)  
Dec 2024
- Trained a 2D bipedal robot in OpenAI Gym (BipedalWalker-v3) to navigate complex terrain through deep reinforcement learning.
  - Designed custom reward shaping combining speed, distance, and joint stability with a step-based delay threshold, driving a 3x increase in distance and velocity over baseline Advantage Actor-Critic (A2C).
  - Implemented Proximal Policy Optimization (PPO) to mitigate policy collapse, boosting average episode moving distance by 132%.

## PROFESSIONAL EXPERIENCE

- Feb 2026 – **AI Software Engineer** (Master's Thesis Internship) | *Viridien*, Massy, France  
Jul 2026 *R&D Department – Seismic Imaging*
- Conducted applied AI research to architect an enterprise advanced RAG system (LlamaIndex, Python), parsing complex geophysical tomography documentation and DSLs to boost response accuracy from 54% to 90%.
  - Optimized ingestion and retrieval via hierarchical chunking, LLM metadata extraction, hybrid search (dense + BM25), and reranking.
  - Benchmarked multi-variable configurations (chunk sizes, embedding models, fusion strategies, LLMs, and top-k) to systematically double retrieval efficiency.
  - Engineered a Model Context Protocol (FastMCP) server exposing 5 custom tools for database retrieval and agent prompt skills.
  - Containerized production infrastructure for the official enterprise AI platform using Docker, Qdrant, MongoDB, and Ollama.

- Sep 2025 – **Human-Robot Interaction Developer** (Corporate Academic Project) | *Air France Industries*, Orly, France  
Jan 2026
- Deployed autonomous visitor onboarding and multimodal interaction flows on the Pepper humanoid robot for the Air France Roissy MRO Lab by engineering native Android applications (Kotlin, SoftBank QiSDK).
  - Reverse-engineered legacy robot architecture and internal systems to build a real-time engagement pipeline combining sensor-based presence detection, voice commands, and tablet UI.
  - Modernized the robot's dialogue pipeline by integrating Mistral LLM APIs, upgrading basic command handling into dynamic, context-aware conversational flows.
- Jul 2025 – **Backend & Cloud Developer** (Freelance) | *EPF Projets*, Cachan, France  
Oct 2025
- Engineered the backend for "Carbon Track", a carbon-footprint mobile application, building REST APIs (Node.js, Express) connected to ADEME and Open Food Facts environmental databases.
  - Designed a hybrid database & serverless backend combining Firebase (Cloud Functions, Firestore, Auth) with local SQLite for offline data persistence.
  - Automated mobile build and release pipelines using the Expo/EAS CI/CD stack to test, package, and deploy the native application onto the Google Play Store.

## TECHNICAL SKILLS

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- **Programming:** Python (Pytorch), Java, SQL
- **Transformer & LLM Architectures:** Multi-Head & Causal Attention, Positional Embeddings, KV-Caching, SFT, LoRA
- **Reinforcement Learning & Control:** MDP, Q-Learning, A2C, PPO, Custom Reward Shaping, Gymnasium / OpenAI Gym
- **GenAI & Information Retrieval:** Advanced RAG (Hierarchical Chunking, Hybrid Search, Reranking), MCP, LlamaIndex, LangChain, Hugging Face, Vector Databases (Qdrant), Ollama
- **Systems, Tools & Infrastructure:** Linux/Bash, Docker, Git, CI/CD, MongoDB, Cloud Platforms (AWS, Azure, GCP)
- **Languages:** English (Professional, TOEIC 895), French (Native), Korean (Basic, SKKU Coursework)